

A Spectral Convolutional Net for Co-Optimization of Integrated Voltage Regulators and Embedded Inductors

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Abstract—Integrated voltage regulators (IVR) with embedded inductors is an emerging technology that provides point-of-load voltage regulation to high-performance systems. Conventional two-step approaches to the design of IVRs can suffer from sub-optimal design as the optimal inductor depends on the characteristics of the buck converter (BC). Furthermore, inductor-level trade-offs such as AC and DC resistance, inductance and area can not be determined independently from the BC. This co-dependency of the BC and the inductor creates a highly non-linear response surface, which raises the necessity of co-optimization, involving multiple time-consuming electromagnetics (EM) simulations.

In this paper, we propose a machine learning based optimization methodology that eliminates EM simulations from the optimization loop to significantly reduce the optimization complexity. A novel technique named as Spectral Transposed Convolutional Neural Network (S-TCNN) is presented to derive an accurate predictive model of the inductor frequency response using a small amount of training data. The derived S-TCNN is then used along with a time-domain model of the BC to perform multi-objective optimization that approximates the Pareto front for 5 objectives, namely inductor area, BC settling time, voltage conversion efficiency, droop and ripple. The resulting methodology provides multiple Pareto optimal inductors in an efficient and fully automated fashion, thereby allows to rapidly determine the optimal trade-offs for possibly contradicting design objectives. We demonstrate the proposed framework on co-optimization of solenoidal inductor with magnetic core and BC that are integrated on silicon interposer.

Keywords—convolutional networks, integrated voltage regulators, embedded inductors, system-level optimization.

I. INTRODUCTION

From automotive to aerospace, computer server systems to consumer products, high-performance electronic systems are realized through high-density integration of digital, analog and mixed-signal ICs into a multi-functional chip, package or interposer. As the level of integration raises, however, energy-efficient operation of such multi-functional devices require dynamic voltage and frequency scaling (DVFS) at different voltage levels to operate at higher performances [1]. This poses challenges to power delivery architectures that needs to rapidly react to rapidly switching current profile of such ICs. Conventional choice of discrete voltage regulators (VRM) that are integrated on board have limited capability of addressing these challenges due to the parasitics involved in the current

path between the VRM on board and the IC. These parasitics cause a highly resonant impedance profile at higher frequencies that limits the switching frequency of board level VRMs, thus, decreasing their transient performance.

One solution to address such challenges are point-of-load active power regulation through switched-inductor integrated voltage regulators (IVR). Here, the VRM and the passives are integrated on the chip, package or interposer to eliminate the parasitics in the current path, allowing to operate at higher switching frequencies that significantly decreases settling time as the load current changes and provides lower power supply noise. The bottleneck of IVRs are the power inductors that need to have sufficient inductance density to be integrated closer to the IC. For this purpose, solenoidal inductors with magnetic cores are previously used in the literature [2], [3].

Design and optimization of such inductors is a challenging and CPU intensive task due to high number of geometrical parameters that determine the shape of the solenoidal structure. The problem is exacerbated with the use of full-wave electromagnetic (EM) simulations required to accurately characterize the frequency response of the inductance and the effective series resistance (ESR) up to multiple harmonics of the switching frequency. Conventional techniques employ a two-step procedure for the design of inductors. First, the inductor is optimized to maximize inductance density and minimize AC & DC losses. This inductor is then used in a time-domain simulation of the buck converter to determine the optimal switching frequency and the output capacitance to minimize settling time, voltage droop and ripple, while maximizing conversion efficiency. For the inductor optimization stage, closed-form approximations are commonly used to model inductance and ESR to reduce computational time required for the optimization [4]. However, a full-wave simulation that accounts for eddy currents, proximity and demagnetization effect is required for accurate characterization of such inductors [5]. Further, a two-step optimization procedure results in sub-optimal designs as the inductor level trade-offs regarding AC & DC losses, inductance and area can not be determined independently from the control parameters of the buck converter. This co-dependency of the inductor and the buck converter raises the necessity of a co-optimization framework that involves full-wave EM solvers in the optimization loop. From the optimization perspective, the problem corresponds to a high dimensional, non-convex, black-box, multi-objective optimization with CPU intensive

function evaluations.

The utilization of EM simulations limits the number of function queries that can be performed during the optimization and creates a bottleneck in the co-optimization framework. A promising approach to bypass EM solvers without giving up the accuracy is machine learning based optimization. Here, a small amount of EM simulations are used to train a learning-based model, such as fully-connected neural networks (FC-NN), which can be as accurate as the EM solver and is fast to evaluate. These methodologies have been proven to work well in the literature for predicting several output design metrics [6]. However, many problems in the interest of EDA community, including the IVR co-optimization problem, involve learning the whole frequency response rather than several output metrics. This makes every frequency point a separate output dimension that needs to be learned, thereby exponentially increasing the number of free parameters, i.e. weights in a FC-NN, to be optimized during the training process, which makes the model prone to overfitting. A common approach to address this problem is using a transfer function (TF) representation of the frequency response using the Vector Fitting algorithm [7] and to learn the mapping from input parameters to the coefficients of the TF [8], [9]. However, the coefficients of the TF have a very wide and non-linear spread when the input sample space is large, which increases the complexity of the problem. More recent approaches use TF method for pre-training the network and use an FC-NN for the detailed training [10], which still is impacted by the aforementioned drawbacks of the FC-NN.

In this paper, we address this problem and propose a new model, named as Spectral Transposed Convolutional Neural Network (S-TCNN), to learn the frequency response of electronic devices by utilizing 1D convolution operations. The number of free parameters and the training complexity of learning based models can be greatly reduced by exploiting the structure of the data. In the case of frequency response of electronic devices, the data structure we have is the spatial correlation along the frequency spectrum, i.e. high correlation between the neighbouring frequency points. Here, we exploit this spatial correlation structure using convolution operations to learn the whole frequency response with a reduced number of learnable parameters in the training phase. The proposed S-TCNN results in a model with reduced training complexity that better generalizes to the cases outside of the training data as compared to the FC-NN counterparts. Further, we present a new loss function to replace the commonly used mean-squared error (MSE) for learning the frequency response. Here, the loss function we propose aims to minimize the loss between the predicted and the actual frequency response as a whole rather than the average loss at individual frequency points as in the case of MSE loss. To the best of our knowledge, approaches using convolutional neural networks have not been explored in the tasks of learning frequency responses of electronic devices in the open literature.

To demonstrate the effectiveness of the proposed method, the presented S-TCNN model is used to accurately learn the frequency response of inductance and ESR of a solenoidal inductor with magnetic core using a small number of training

data. The model is then used along with a time-domain Simulink model of the IVR for multi-objective co-optimization of IVR and the inductor with the purpose of approximating the Pareto front of 5 objectives, namely conversion efficiency, settling time, voltage droop, ripple and inductor area.

The rest of this paper is structured as follows: Section II presents the proposed S-TCNN model to learn the frequency response along with the proposed training loss function; Section III presents the application of the proposed method to derive the model of solenoidal inductor with magnetic core; Section IV provides the IVR and inductor co-optimization framework followed by the conclusion in Section V.

II. CONVOLUTIONAL NEURAL NETWORKS FOR LEARNING FREQUENCY RESPONSE

A. Convolutional Layers

A convolutional layer in neural networks aims to learn local patterns from the input data. In the context of dealing with frequency responses of electronic devices, this corresponds to searching for patterns such as resonances, ripples and flat regions in smaller frequency bands. This is achieved by employing a sliding inner product operation on small frequency bands and sharing the learnable weights in the hidden units across the whole frequency spectrum, which results in a reduced number of weights that need to be optimized during the training. Most commonly used form of CNNs utilize a 2D sliding product to consider spatial correlation in two axis of the data, such as the width and height of an image. For the cases of frequency responses, the spatial correlation to be exploited is along the frequency axis. Hence, a 1D operation needs to be utilized. The 1D sliding inner product corresponds to a cross-correlation operation between the hidden units, referred as a 1D kernel, and the inputs to the convolutional layer, given as:

$$y = f(\mathbf{h} * \mathbf{x}) = f \left(\sum_{m=-\infty}^{m=\infty} x[n]h[m-n] \right) \quad (1)$$

where $(*)$ denotes the cross-correlation operation, $f(\cdot)$ is the non-linear activation function, h is the kernel that contains the learnable weights, x and y are the input and the output of the convolutional layer, respectively. Depending on the smoothness of the response, the number of kernels can be increased to learn more distinct patterns. As the data is passed deeper into the network hierarchy through non-linear activation functions, the patterns become more and more abstract to describe the low-level details of the response such as a rapid change from flat response to resonances.

The limits of the sum in (1) depend on the size of $\mathbf{x}$ and $\mathbf{h}$ and whether a padding is used or not to account for non-overlapping parts of these vectors, which also determines the size of the output vector $\mathbf{y}$. As we are interested in patterns in smaller frequency bands, the size of $\mathbf{h}$ will be much smaller than the size of $\mathbf{x}$. As the small sized $\mathbf{h}$ slides over $\mathbf{x}$, handling the cases at the edges, i.e. non-overlapping parts, becomes critical. Commonly used zero-padding the input technique corresponds to creating a sharp drop to zero at the start & end of the frequency response and disturbs the spatial correlation.

Hence, the so-called *valid convolution* operation should be employed where only the overlapped portions of the $\mathbf{x}$ & $\mathbf{h}$ are preserved in the output to avoid such sharp drops. Note that a valid-convolution results in a learnable downsampling operation where the output contains a filtered, compressed version of the input.

B. Transposed Convolutional Layer

Cross-correlation operation in (1) is very useful for extracting patterns from the frequency response. However, our goal in this paper is predicting the frequency response rather than extracting patterns from it. Thus, the flow in the network is from input parameters such as geometry and/or material properties to the high-dimensional frequency response rather than the opposite direction as with the convolutional layers. The flow can be reversed by making use of *transposed convolutional layers*. A transposed convolutional layer is a learnable upsampling method that preserves the spatial correlation in its input by considering a particular input as the result of cross-correlation operation between the output and the learnable kernel.

This is best demonstrated when the sliding inner product is unrolled as a Toeplitz matrix [11]. Consider the case of a single layer 1D CNN to which the input is the frequency response of a device at m frequency points, i.e. $\mathbf{x} = [x(f_1), x(f_2), \dots, x(f_m)]^T$. The output $\mathbf{y}$ then can be written as the cross-correlation of x with a kernel $\mathbf{h} = [w_1, w_2, \dots, w_k]$ as the following matrix multiplication:

$$\mathbf{y} = f(\mathbf{h} * \mathbf{x}) = f(H\mathbf{x}) \quad (2)$$

with

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}, H = \begin{bmatrix} w_1 & w_2 & \dots & w_k & 0 & 0 & 0 & \dots & 0 \\ 0 & w_1 & w_2 & \dots & w_k & 0 & 0 & \dots & 0 \\ 0 & 0 & w_1 & w_2 & \dots & w_k & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & 0 & 0 & 0 & w_1 & \dots & w_{k-1} & w_k \\ 0 & 0 & 0 & 0 & 0 & w_1 & \dots & w_{k-1} & w_k \end{bmatrix}, \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_m \end{bmatrix} \quad (3)$$

where $n = m - k + 1$ is the resulting downsampled output vector y . To reverse this operation, the transposed convolutional operation can be written as:

$$\hat{\mathbf{x}} = f(\hat{\mathbf{h}} * \hat{\mathbf{y}}) = f(\hat{H}^T \hat{\mathbf{y}}) \quad (4)$$

with

$$\hat{\mathbf{x}} = \begin{bmatrix} \hat{x}_1 \\ \hat{x}_2 \\ \vdots \\ \hat{x}_m \end{bmatrix}, \hat{H}^T = \begin{bmatrix} w_1 & 0 & \dots & 0 & 0 \\ w_2 & w_1 & \dots & \vdots & \vdots \\ \vdots & w_2 & \dots & 0 & 0 \\ w_{k-1} & \vdots & \dots & w_1 & 0 \\ w_k & w_{k-1} & \dots & w_2 & w_1 \\ 0 & w_k & \dots & \vdots & w_2 \\ 0 & 0 & \dots & w_{k-1} & \vdots \\ \vdots & \vdots & \dots & w_k & w_{k-1} \\ 0 & 0 & \dots & 0 & w_k \end{bmatrix}, \hat{\mathbf{y}} = \begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \vdots \\ \hat{y}_n \end{bmatrix} \quad (5)$$

where $m = n + k - 1$ is the resulting upsampled output vector y . The H and H^T matrices in (3) and (5) also show how the parameter sharing phenomena works by exploiting the spatial correlation in the frequency axis that reduces the training complexity. Fig. 1 further illustrates and compares the operations done by and the flow of data in convolutional and transposed convolutional layers.

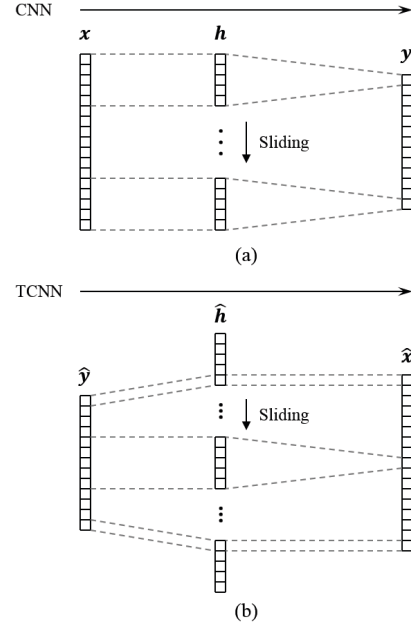


Fig. 1: Comparison of the operations done by CNN and TCNN

C. Removing the Translational Invariance

The sliding inner product, along with the downsampling operations, results in a transformation that is translationally invariant. That is, the output of a cross-correlation in (1) will have a high value if a certain pattern is present regardless of its location at the input vector. For image classification problems where CNNs shine at, this is a highly desirable property since the network can classify the image based on the presence of the object rather than focusing on which pixel contains it.

For the case of learning frequency responses, this means that the TCNN model is able to learn the presence of ripples or resonances in small frequency bands, but can struggle when distinguishing the location of two similar patterns in the frequency spectrum, which is not a desirable property. In order to address this issue, we use a *CoordConv* layer as the last layer of the network. *CoordConv* layer [12] is a recently developed method that simply hard-codes “where” information to the output by concatenating a coordinate axis to its input. This is done by concatenating the frequency points, f_1 through f_k , as an additional input to the last layer of the S-TCNN architecture. Since the weights that are multiplied by this concatenated frequency axis can be adjusted to ignore this operation, we explicitly let the network to have the ability to learn if the translational invariance property should be kept or discarded from the frequency axis of the S-TCNN model.

The final S-TCNN network architecture we propose for predicting the frequency responses from a set of control parameters is given in Fig. 2. Here, the input vector goes through a set of learnable and non-linear upsampling operations to construct the high-dimensional frequency response at k frequency points

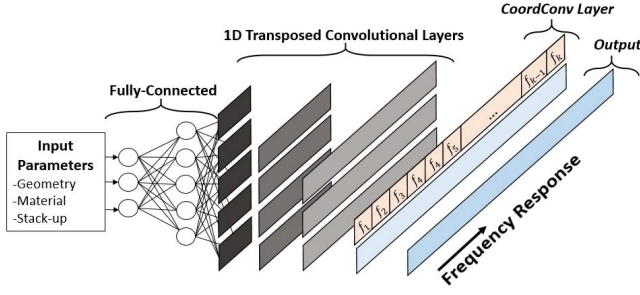


Fig. 2: The proposed S-TCNN architecture for mapping input parameters predicting frequency response.

by preserving and exploiting the spatial correlation in the frequency axis, while the last layer hard-codes the location information to remove the translational invariance.

D. Loss Function for Frequency Responses

One of the most commonly used loss functions for training a TCNN, or any other network architecture, is the mean squared error, given as:

$$L = \frac{1}{NK} \sum_{n=1}^N \sum_{k=1}^K (y_{n,k} - \hat{y}_{n,k})^2 \quad (6)$$

where $\hat{y}_{n,k}$ is the k^{th} dimension of the output of the network at n^{th} training sample. As we consider each frequency point as a separate output dimension, L in (6) corresponds to minimizing the reconstruction loss at individual frequency points.

However, we want the S-TCNN model to learn weights to minimize the reconstruction error of the frequency response as a whole rather than at individual points. Here, we propose a modified loss function to change the behavior of the S-TCNN model to reconstruct the frequency response, which can be written as:

$$L_{freq} = \frac{1}{N} \sum_{n=1}^N \sqrt{\frac{1}{K} \sum_{k=1}^K (y_{n,k} - \hat{y}_{n,k})^2} \quad (7)$$

The loss function we propose in (7) is based on the scaled ℓ^2 -norm of the error between the predicted and the actual frequency response, averaged over N different frequency responses in the training set. This ensures that the model learns the mapping from a certain input vector to the frequency response as a whole, rather than the mapping to k different frequency points. As we show in the subsequent section, the L_{freq} loss in (7) improves the accuracy of the network not just for the S-TCNN model, but also for the regular fully-connected networks for the task of predicting frequency responses.

III. SOLENOIDAL INDUCTOR MODEL

A. Simulation Setup

The solenoidal inductor considered in this work uses a Nickel-Zinc (NiZn) ferrite magnetic core and is integrated

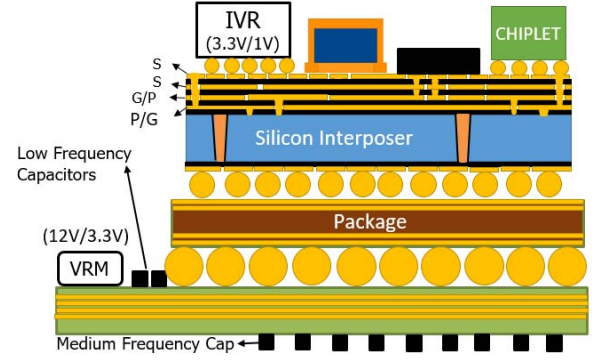


Fig. 3: Stack-up of the considered 2.5D integrated system.

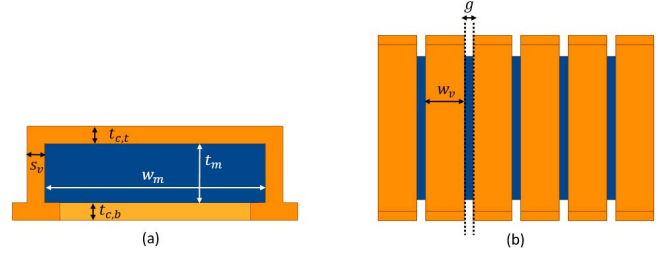


Fig. 4: Geometry of the solenoidal inductor. (a) Side view. (b) Top view.

TABLE I: Control Parameters of Solenoidal Inductor

Parameter	Unit	Min	Max
Gap between windings	g mil	2	20
Number of windings	N	3	13
Size of via	s _v μm	50	103
Copper Trace Width	w _c mil	2	20
Copper Thickness Bottom	t _{c,b} μm	35	170
Copper Thickness Top	t _{c,t} μm	35	170
Magnetic Core Thickness	t _d μm	50	650
Magnetic Core Width	w _d μm	50	350

on the top metal layer of a silicon interposer based 2.5D heterogeneously integrated system as in Fig. 3. Its geometry is defined by eight parameters as shown in Fig. 4 and the corresponding bounds for each parameter are given in Table I.

In order to create the predictive model of the solenoidal inductor, 1000 samples based on Latin Hypercube Sampling (LHS) are determined. Measured complex permeability of the NiZn core [13] is then imported into a full-wave EM solver, Ansys HFSS [14], to simulate the 2-port Y-parameters at 200 frequency points between 10 MHz and 500 MHz. The Y-Parameters are then converted to a pi-equivalent circuit to extract the frequency dependent inductance and effective series

resistance, i.e. $L(f)$ and $ESR(f)$. Note that the pi-equivalent circuit here is not an approximation since any reciprocal 2-port device can be represented by frequency dependent passive elements [15].

Once the data collection is completed, 800 out of 1000 samples are standardized to have zero-mean and unit standard deviation, and used as the training data for the S-TCNN model that maps 8 input parameters to $L(f)$ and $ESR(f)$.

B. Results

We compare the proposed S-TCNN model with the FC-NN and train both models using both MSE loss in (6) and the proposed L_{freq} loss in (7). To assess the quality of models, we use normalized mean-squared error (NMSE) averaged over each frequency response in the validation set, given as:

$$NMSE = \frac{1}{N} \sum_{n=1}^N \left(\frac{\sum_{k=1}^K (y_{n,k} - \hat{y}_{n,k})^2}{\sum_{k=1}^K (y_{n,k} - \frac{1}{K} \sum_{k=1}^K y_{n,k})^2} \right) \quad (8)$$

where N is the size of the validation set and k is the frequency axis. We choose NMSE metric to evaluate the models as it is highly intuitive and provides a normalized scale for different outputs, i.e. $L(f)$ & $ESR(f)$. Here, an NMSE value of 1.0 means that the model is able to predict no better than the mean of each frequency response.

Table II summarizes NMSE values for each model and training loss function. The proposed S-TCNN model results in 10.8% improvement in prediction accuracy as compared to the most commonly used architecture in the literature, FC-NN model that is trained on MSE loss. The use of the proposed L_{freq} loss function resulted in 3.2% and 5.1% improvement in prediction accuracy for FC-NN and S-TCNN models that are trained using the MSE loss, showing its efficacy for predicting frequency responses. This is further demonstrated in Fig. 6. Here, it can be seen that in addition to the final value, the convergence of validation NMSE is faster and more robust for both FC-NN and S-TCNN when trained with L_{freq} loss. This shows that the proposed loss function reduces the training complexity and better represents the generalization capability of the network to unseen frequency responses as compared to the training with MSE loss.

TABLE II: Model Comparison

	FC-NN		Proposed S-TCNN	
	MSE	L_{freq}	MSE	L_{freq}
Validation NMSE	0.228	0.177	0.152	0.120
Run Time	0.01 sec		1.503 sec	
<i>(for 1k freq. responses)</i>	<i>(HFSS: ~ 25 hours)</i>			

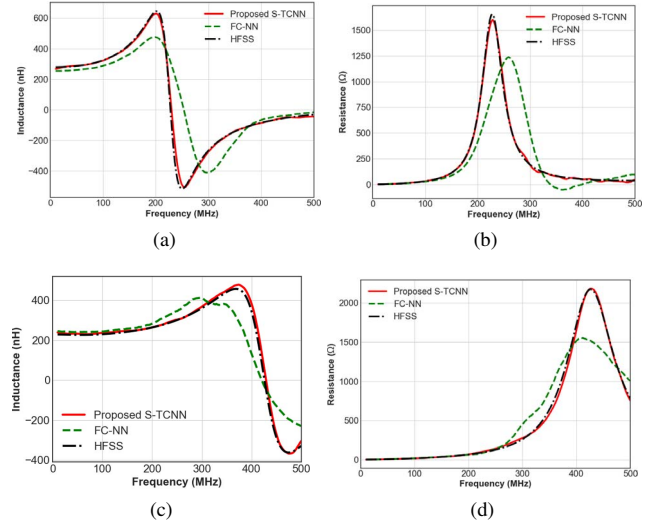


Fig. 5: Comparison of S-TCNN & FC-NN to EM simulations for two test cases. (a), (b) Test case #1. (c), (d) Test case #2.

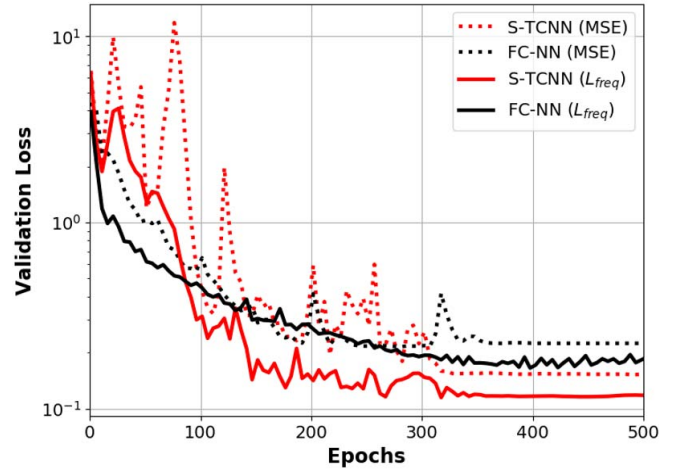


Fig. 6: Convergence of validation loss for different models.

When both models are trained with the L_{freq} loss, S-TCNN showed 5.2% increased accuracy compared to FC-NN, which shows its ability to exploit the spatial correlation in the frequency axis to learn the patterns of $L(f)$ & $ESR(f)$. As further illustrated in Fig. 5, the predictions done by S-TCNN capture the behavior of self-resonance patterns seen in $L(f)$ & $ESR(f)$ very accurately, whereas FC-NN shows poor performance for learning this pattern.

In terms of the run times, S-TCNN took 1.503 sec to generate frequency responses of 1k different designs as compared to 25 hours by HFSS. FC-NN was slightly faster as it took 0.01 sec for the same task, however, the difference is insignificant for the purposes of optimization.

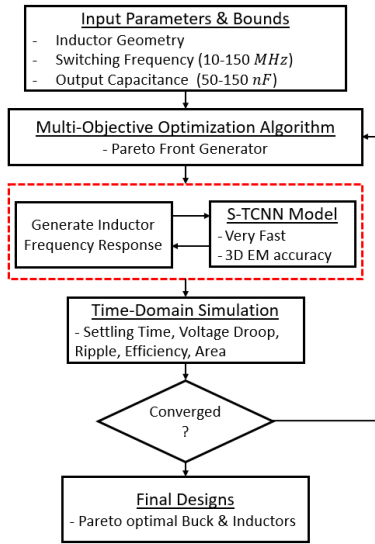


Fig. 7: IVR & Inductor co-optimization flow using the proposed S-TCNN model.

Note that for a fair comparison, hyperparameters of both FC-NN and S-TCNN, such as learning rate and network architecture, are optimized using grid-search. The resulting S-TCNN architecture had 13509 trainable parameters and consisted of 3 fully-connected layers, each having 25 hidden units, followed by 3 transposed convolutional layers, each having 10 output channels. The kernel size and stride for these layers were 31, 31, 20 and 1, 2 and 2, respectively. The best performing FC-NN architecture had 45950 trainable parameters and consisted of 3 layers that have 50, 100 and 400 hidden units, respectively. Both models had batch normalization in between layers and used tanh as the activation function. The training was performed in PyTorch [16] using Adam optimizer [17] with a learning rate of 0.01.

IV. IVR & INDUCTOR CO-OPTIMIZATION

Electrical characteristics of the power inductor directly affect both transient performance and conversion efficiency of the final IVR design. As the IVR chip and the inductor are to be integrated on the interposer, the inductance density needs to be sufficiently high to enable this integration on a feasible area. The switching frequency of the IVR needs to be increased to reduce the inductor area, voltage ripple and settling time, which, however, significantly increases switching losses as well as the inductor AC losses as ESR increases with the switching frequency and at its harmonics. The output capacitance of the IVR should be increased to minimize voltage droops and ripple, which results in increased settling time. The optimal output capacitance would then be the minimum value that achieves droop and ripple specifications. As these specifications also depend on the inductance and the switching frequency, the capacitance needs to be considered as a parameter in the co-optimization framework.

In order to find the optimal IVR & inductor design, the optimal trade-offs between settling time, voltage droop, ripple,

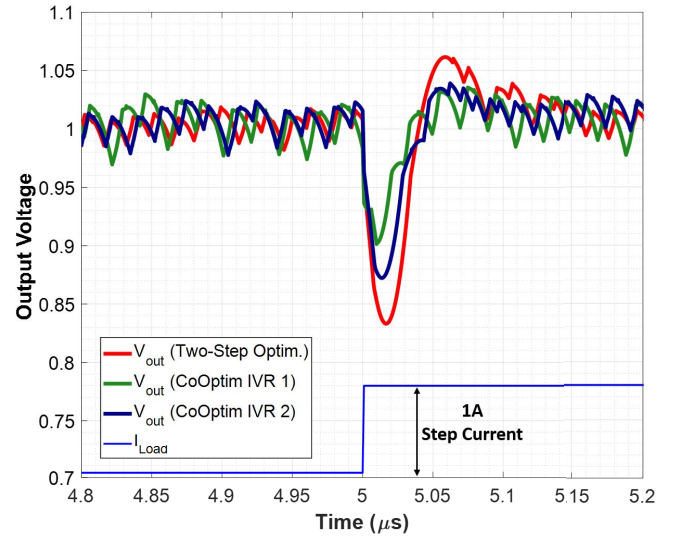


Fig. 8: Comparison of transient response of optimized IVRs to a 1A step current.

TABLE III: Comparison of Optimized IVRs

	Two-Step Optimization	Co-Optimized IVR 1	Co-Optimized IVR 2
Switching Freq.	125 MHz	100 MHz	115 MHz
Capacitance	100 nF	115 nF	128 nF
Inductance	29.8 nH	20.7 nH	23.8 nH
ESR	3.63 Ω	1.01 Ω	1.12 Ω
DC Resistance	10.5 m Ω	15.7 m Ω	30.2 m Ω
Area	5.12 mm ²	4.64 mm ²	2.48 mm ²
Efficiency	76.6 %	77.8 %	76.3 %
Voltage Droop	167 mV	98.6 mV	127 mV
Voltage Ripple	38.8 mV	49.3 mV	40.2 mV
Settling Time	115 ns	80 ns	75 ns

conversion efficiency and inductor area need to be determined. This raises the necessity to perform a multi-objective co-optimization to approximate the Pareto front of these objectives instead of a weighted sum approach that combines them into a single-objective. The use of Pareto front generates multiple non-dominated IVR & inductor designs, meaning that each design can not be further improved for one objective without degrading the others. Different non-dominated designs can then be selected to prioritize one objective over another based on design specifications.

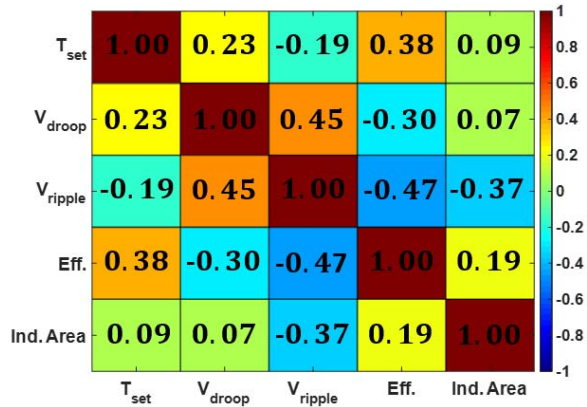


Fig. 9: Correlation matrix of the Pareto optimal designs.

A. Optimization Setup

The optimization setup used for the IVR & inductor co-optimization is given in Fig. 7. Here, the geometrical parameters of the inductor along with switching frequency and output capacitance value are chosen by a multi-objective optimization algorithm and fed into the previously derived S-TCNN model to generate the inductor frequency response. $L(f)$ & $ESR(f)$, along with the PDN impedance, are then fed into a Simulink model of the buck converter to perform time-domain simulation to extract settling time, droop and ripple for a for 3.3V/1V voltage conversion with 1A step current as the load. Finally, the voltage conversion efficiency is calculated using a comprehensive model that accounts for switching and conduction losses of power switches, DC and AC losses of inductor, PDN and output capacitance [3].

Note the use of the S-TCNN model for characterizing the inductor results in very short simulation times and allows for querying the objective function many times. Hence, any multi-objective optimization algorithm can be used in the co-optimization flow in Fig. 7. In this paper, we choose the popular method of NSGA-II as it has shown to be successful in previous EDA problems [18].

B. Results

We compare our results with the IVR design that is optimized in two-steps: 1) a thorough optimization of the inductor for maximum conversion efficiency and minimum area [5], and 2) optimization of switching frequency and output capacitance to maximize transient performance. We choose 2 designs from the resulting Pareto front of the proposed approach, IVR1 and IVR2, that prioritizes transient performance and area, respectively, while maintaining a voltage ripple constraint of 50 mV.

The resulting 5-dimensional Pareto front is given in Fig. 10, showing the comparison of 105 different Pareto optimal IVR & inductor designs to illustrate the optimal design trade-offs. The inductor area has shown a negative correlation with the voltage ripple and positive correlation with the conversion efficiency, meaning that they can not be improved at the same time

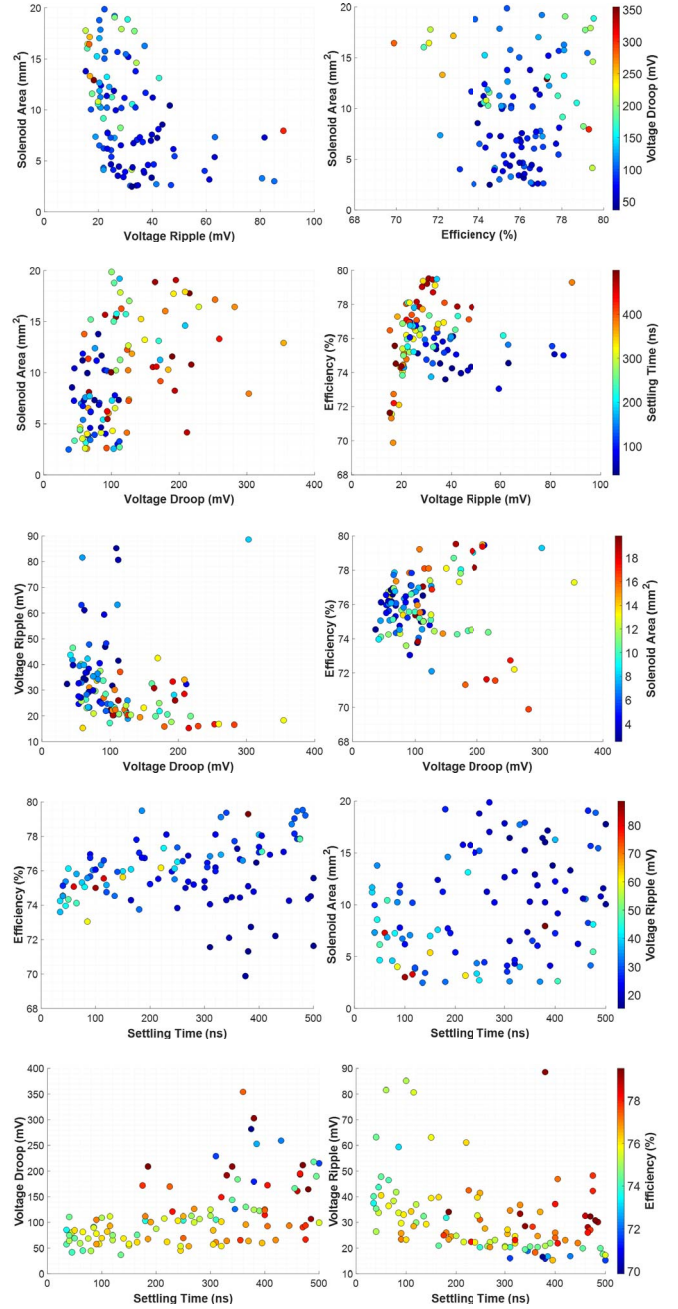


Fig. 10: 5 dimensional Pareto front showing optimal trade-offs for IVR & inductor co-design.

while maintaining Pareto optimality. Further, the conversion efficiency has shown a strong positive correlation with the settling time. These suggest that between any two non-dominant designs, the efficiency can not be improved without degrading the transient performance and the solenoidal inductor, within the sample space, can not be further miniaturized without degrading the efficiency and the transient performance. The

correlation matrix between the Pareto optimal designs is also given in Fig. 9 to further illustrate the optimal trade-offs for co-design of the IVR & solenoidal inductor with NiZn magnetic core.

The electrical characteristics of the co-optimized IVRs are given in Table III and in Fig. 8 and compared to the resulting design from a two-step optimization procedure. The design that prioritized inductor miniaturization, IVR2, has 51.56% and 46.5% reduced area compared to the two-step optimized design and IVR1, respectively. As the miniaturization results in decreased copper trace widths, the DC resistance is significantly increased compared to other designs, resulting in reduced conversion efficiency of 76.3%. On the other hand, IVR1 has the highest efficiency with 77.8%, along with 40.9% and 22.3% improved voltage droop as compared to two-step optimization and IVR2, respectively. Comparisons shown in Table III, along with Pareto front in Fig. 10, show that a significant performance improvement in all objectives can be achieved by going through the proposed co-optimization framework.

V. CONCLUSION

In this paper, we have proposed a new method to learn the non-linear mapping from control parameters of an electronic device to its frequency response, named as Spectral Transposed Convolutional Neural Network (S-TCNN). Unlike previous methods in the literature, we've shown that the spatial correlation of the frequency spectrum can be exploited by using transposed convolutional layers to reduce number of learnable parameters in the model, thus, reducing the training complexity. Further, we have proposed a new loss function that is better suited to the reconstruction of frequency response as a whole rather than at individual points. For the application to solenoidal inductor with magnetic core, the proposed S-TCNN model showed 10.8% improvement in validation loss compared to the commonly used fully-connected networks, while the proposed loss function alone provided 5.1% improvement for the FC-NN model and 3.2% improvement for the S-TCNN model.

Moreover, we have used the derived S-TCNN model of the solenoidal inductor to perform multi-objective co-optimization of IVR & inductor. We have shown that how Pareto front of the voltage settling time, droop, ripple, conversion efficiency and inductor area can be estimated in an accurate yet efficient way. Performance of the designs obtained from the Pareto front are compared to the design resulting from a two-step optimization procedure, where up to 51.5%, 40.9% and 26.1% improvements in inductor area, voltage droop and settling time are achieved.

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